



9025 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Cycle: 4, Proposal Category: GO

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JWST Proposal 9025 (Created: Wednesday, July 16, 2025, 7:00:13PM Eastern Standard Time) - Overview

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
WASP 25b				
	1	Transit	NIRSpec Bright Object Time Series	(1) WASP-25
K2-232b				
	2	Transit	NIRSpec Bright Object Time Series	(2) K2-232
WASP 84b				
	3	Transit	NIRSpec Bright Object Time Series	(3) WASP-84
TOI-2031b				
	4	Transit	NIRSpec Bright Object Time Series	(4) TOI-2031
TOI-677b				
	5	Transit 1	NIRSpec Bright Object Time Series	(5) TOI-677
	6	Transit 2	NIRSpec Bright Object Time Series	(5) TOI-677

ABSTRACT

The formation and inward migration of short-period giant exoplanets have remained mysterious. Two migration pathways have been proposed: disk migration and high-eccentricity (high-e) migration. Disk migration is predicted to yield planets in spin-orbit alignment with their host stars, while high-e migration should lead to more diverse inclinations. At the same time, the two migration pathways should also impact atmospheric

compositions in different ways, with disk migrated planets being able to accrete inner disk material while high-e migrated planets cannot, leading to significantly different metallicities and C/O ratios. Both migration pathways are likely important given the diversity in spin-orbit alignments of Hot Jupiters around hot stars above the Kraft Break and the existence of Hot Jupiters in compact multi-planet systems. Recent observations have shown that Warm Jupiters, unlike Hot Jupiters, are all spin-orbit aligned no matter their host star effective temperature, suggesting that they all migrated via the disk. We propose to observe 5 spin-orbit aligned Warm Jupiters with JWST NIRSpec/G395H and combine the data with those of a 6th, previously observed Warm Jupiter, WASP-80 b, to constrain the population level metallicity and C/O ratio. We will then compare our results to those of the Cycle 2 BOWIE-ALIGN program, which observed spin-orbit aligned and misaligned Hot Jupiters orbiting hot stars, and additional JWST programs that targeted aligned, cool star Hot Jupiters, to uncover the link between migration and atmospheric composition of giant exoplanets.

OBSERVING DESCRIPTION

We will observe the transits of 5 warm Jupiters. All observations will be done using NIRSpec/G395H in TSO. The visits will be centered around the mid-transit time of each planet with 1:1 baseline coverage. Target acquisition will be done on the targets themselves for 2 of the planets, while for the remaining 3, which are too bright, nearby target acquisition stars are indicated. All observations will use subarray SUB2048 and will have >10 groups per integration, staying safely below saturation.

Proposal 9025 - Targets - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WASP-25	RA: 13 01 26.3760 (195.3599000d) Dec: -27 31 19.92 (-27.52220d) Equinox: J2000	Proper Motion RA: -29.264 mas/yr Proper Motion Dec: -6.1960000266481075 mas/yr Parallax: 0.0046501" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Jmag = 10.596</i> <i>Category=Star</i> <i>Description=[Exoplanet Systems]</i> <i>Extended=NO</i>				
	(2)	K2-232	RA: 04 55 3.9626 (73.7665108d) Dec: +18 39 16.31 (18.65453d) Equinox: J2000	Proper Motion RA: 61.996 mas/yr Proper Motion Dec: -48.14100002477062 mas/yr Parallax: 0.0077059" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Exoplanet Systems]</i> <i>Extended=NO</i>				
Fixed Targets	(3)	WASP-84	RA: 08 44 25.7032 (131.1070967d) Dec: +01 51 36.11 (1.86003d) Equinox: J2000	Proper Motion RA: -23.344 mas/yr Proper Motion Dec: -31.56999996463128 mas/yr Parallax: 0.0099636" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Exoplanet Systems]</i> <i>Extended=NO</i>				
	(4)	TOI-2031	RA: 22 04 28.0921 (331.1170504d) Dec: +81 33 57.39 (81.56594d) Equinox: J2000	Proper Motion RA: 25.79 mas/yr Proper Motion Dec: 2.398 mas/yr Parallax: 0.0035797" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Jmag = 10.312</i> <i>Category=Star</i> <i>Description=[Exoplanet Systems]</i> <i>Extended=NO</i>				

Proposal 9025 - Targets - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

(5)	TOI-677	RA: 09 36 28.6865 (144.1195271d) Dec: -50 27 47.80 (-50.46328d) Equinox: J2000	Proper Motion RA: -24.807 mas/yr Proper Motion Dec: 42.562 mas/yr Parallax: 0.0070495" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Exoplanet Systems] Extended=NO			
(6)	TA-K2-232	RA: 04 55 2.5336 (73.7605567d) Dec: +18 39 50.50 (18.66403d) Equinox: J2000	Proper Motion RA: 17.530 mas/yr Proper Motion Dec: 16.566 mas/yr Parallax: 0.0013077" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and parameters were updated using GAIA EDR3 information found at:</i> https://vizier.cds.unistra.fr/viz-bin/VizieR-5?-ref=VIZ6877a6712de88c&-out.add=.&-source=I/350/gaiaedr3&-c=073.76063895252%20%2b18.66395491160,eq=ICRS,rs=2&-out.orig=o <i>Jmag = 15.493</i> <i>Kmag = 14.456</i> Category=Star Description=[Exoplanets] Extended=NO			
(7)	TA-WASP-84	RA: 08 44 27.6973 (131.1154054d) Dec: +01 51 1.09 (1.85030d) Equinox: J2000	Proper Motion RA: -6.792 mas/yr Proper Motion Dec: -5.727 mas/yr Parallax: 0.0024860" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and parameters were updated using GAIA EDR3 information found at:</i> https://vizier.cds.unistra.fr/viz-bin/VizieR-5?-ref=VIZ6877ac81b7a6b&-out.add=.&-source=I/350/gaiaedr3&-c=131.11537505935%20%2b01.85027757639,eq=ICRS,rs=2&-out.orig=o <i>Jmag = 16.230</i> <i>Kmag = 15.187</i> Category=Star Description=[Exoplanets] Extended=NO			
(8)	TA-TOI-677	RA: 09 36 27.7478 (144.1156158d) Dec: -50 27 28.44 (-50.45790d) Equinox: J2000	Proper Motion RA: -4.202 mas/yr Proper Motion Dec: 2.677 mas/yr Parallax: 0.0003919" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and parameters were updated using GAIA EDR3 information found at:</i> https://vizier.cds.unistra.fr/viz-bin/VizieR-5?-ref=VIZ6877ac81b7a6b&-out.add=.&-source=I/350/gaiaedr3&-c=144.11558665866%20-50.45788822419,eq=ICRS,rs=2&-out.orig=o <i>Jmag = 12.604</i> <i>Kmag = 12.211</i> Category=Star Description=[Exoplanets] Extended=NO			

Proposal 9025 - Observation 1 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 1: Transit										Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning										
Observing Template: NIRSpec Bright Object Time Series											
Diagnostics	(Transit (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Transit (Obs 1)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition.										
(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	WASP-25	RA: 13 01 26.3760 (195.3599000d) Dec: -27 31 19.92 (-27.52220d) Equinox: J2000			Proper Motion RA: -29.264 mas/yr Proper Motion Dec: -6.1960000266481075 mas/yr Parallax: 0.0046501" Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
Jmag = 10.596 Category=Star Description=[Exoplanet Systems] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	99	272	1	1	272	24539.971		

Proposal 9025 - Observation 1 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9568373605182704 to 0.9679047039751242 with period 3.76483 Days and zero-phase 2455888.66484 HJD Time Series Observation No Parallel Attachments
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Proposal 9025 - Observation 2 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 2: Transit										Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(Transit (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Transit (Obs 2)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition. (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	K2-232	RA: 04 55 3.9626 (73.7665108d) Dec: +18 39 16.31 (18.65453d) Equinox: J2000			Proper Motion RA: 61.996 mas/yr Proper Motion Dec: -48.14100002477062 mas/yr Parallax: 0.0077059" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	6 TA-K2-232	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	20	2091	1	1	2091	39650.546		

Proposal 9025 - Observation 2 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9776155231511899 to 0.9813462692926583 with period 11.168454 Days and zero-phase 2458160.403851 HJD Time Series Observation No Parallel Attachments
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Proposal 9025 - Observation 3 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 3: Transit										Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning										
Observing Template: NIRSspec Bright Object Time Series											
Diagnostics	(Transit (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Transit (Obs 3)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition.										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	WASP-84	RA: 08 44 25.7032 (131.1070967d)			Proper Motion RA: -23.344 mas/yr					
			Dec: +01 51 36.11 (1.86003d)			Proper Motion Dec: -31.56999996463128 mas/yr					
			Equinox: J2000			Parallax: 0.0099636"					
						Epoch of Position: 2000					
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[Exoplanet Systems]										
	Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	7 TA-WASP-84	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	30	876	1	1	876	24512.652		

Proposal 9025 - Observation 3 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9809350540143591 to 0.985823501702985 with period 8.52349648 Days and zero-phase 2456448.052286 HJD Time Series Observation No Parallel Attachments
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Proposal 9025 - Observation 4 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 4: Transit											Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Bright Object Time Series											
Diagnostics	(Transit (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.											
	(Transit (Obs 4)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition.											
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	TOI-2031	RA: 22 04 28.0921 (331.1170504d)				Proper Motion RA: 25.79 mas/yr					
			Dec: +81 33 57.39 (81.56594d)				Proper Motion Dec: 2.398 mas/yr					
			Equinox: J2000				Parallax: 0.0035797"					
							Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Jmag = 10.312											
	Category=Star											
	Description=[Exoplanet Systems]											
	Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08		
Template	Subarray											
	SUB2048											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	G395H/F290LP	NRSRAPID	85	372	1	1	372	28864.403			

Proposal 9025 - Observation 4 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9671943759 to 0.9744845146 with period 5.7154834 Days and zero-phase 2459634.565 HJD Time Series Observation No Parallel Attachments
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Proposal 9025 - Observation 5 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 5: Transit 1										Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(Transit 1 (Obs 5)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Transit 1 (Obs 5)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition.										
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	TOI-677	RA: 09 36 28.6865 (144.1195271d) Dec: -50 27 47.80 (-50.46328d) Equinox: J2000			Proper Motion RA: -24.807 mas/yr Proper Motion Dec: 42.562 mas/yr Parallax: 0.0070495" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	8 TA-TOI-677	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	20	1255	1	1	1255	23797.912		

Proposal 9025 - Observation 5 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9859091421485741 to 0.9896172626357914 with period 11.2366 Days and zero-phase 2458918.28242 HJD Time Series Observation No Parallel Attachments
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Proposal 9025 - Observation 6 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Observation	Proposal 9025, Observation 6: Transit 2										Thu Jul 17 00:00:13 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(Transit 2 (Obs 6)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Transit 2 (Obs 6)) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition.										
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	TOI-677	RA: 09 36 28.6865 (144.1195271d)			Proper Motion RA: -24.807 mas/yr					
			Dec: -50 27 47.80 (-50.46328d)			Proper Motion Dec: 42.562 mas/yr					
			Equinox: J2000			Parallax: 0.0070495"					
						Epoch of Position: 2000					
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[Exoplanet Systems]										
	Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	8 TA-TOI-677	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G395H/F290LP	NRSRAPID	20	1255	1	1	1255	23797.912		

Proposal 9025 - Observation 6 - The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

Special Requirements	Phase 0.9859091421485741 to 0.9896172626357914 with period 11.2366 Days and zero-phase 2458918.28242 HJD Time Series Observation No Parallel Attachments
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